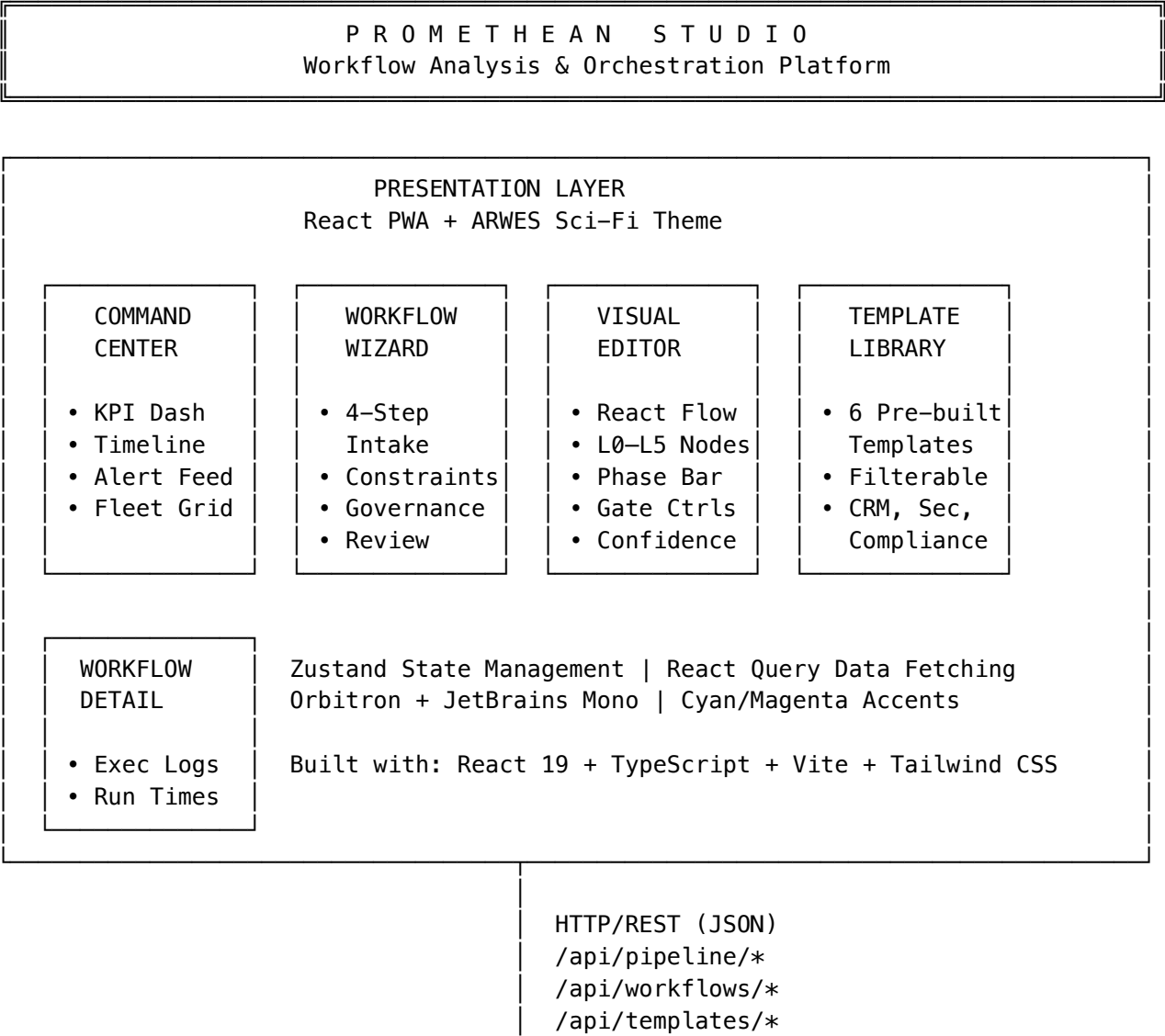


PROMETHEAN — System Architecture Diagram

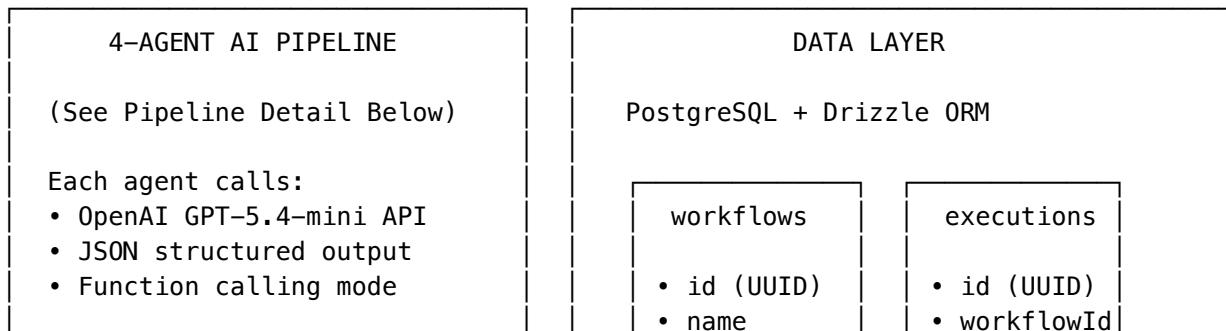
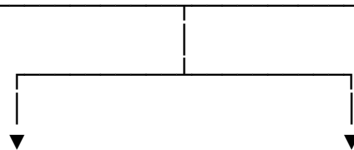
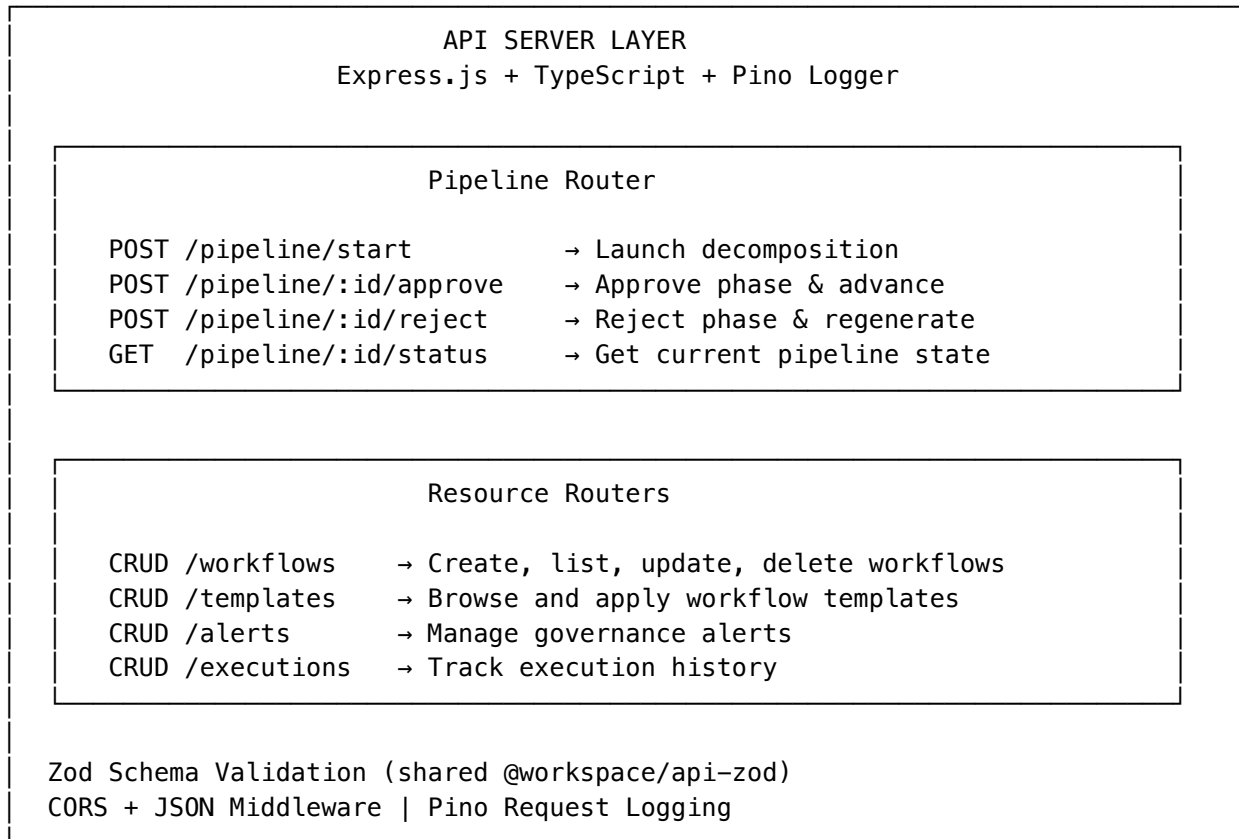
Phase 2 Deliverable | Track A: Technical Build

April 2026

1. High-Level System Architecture



| /api/alerts/*
▼



- nodes[]
- edges[]
- phase
- status
- govConfig

- status
- result
- cost
- latency

templates

- id (UUID)
- name
- nodes[]
- edges[]
- domain
- rating

alerts

- id (UUID)
- workflowId
- alertType
- severity
- evidence
- status

workflow_versions

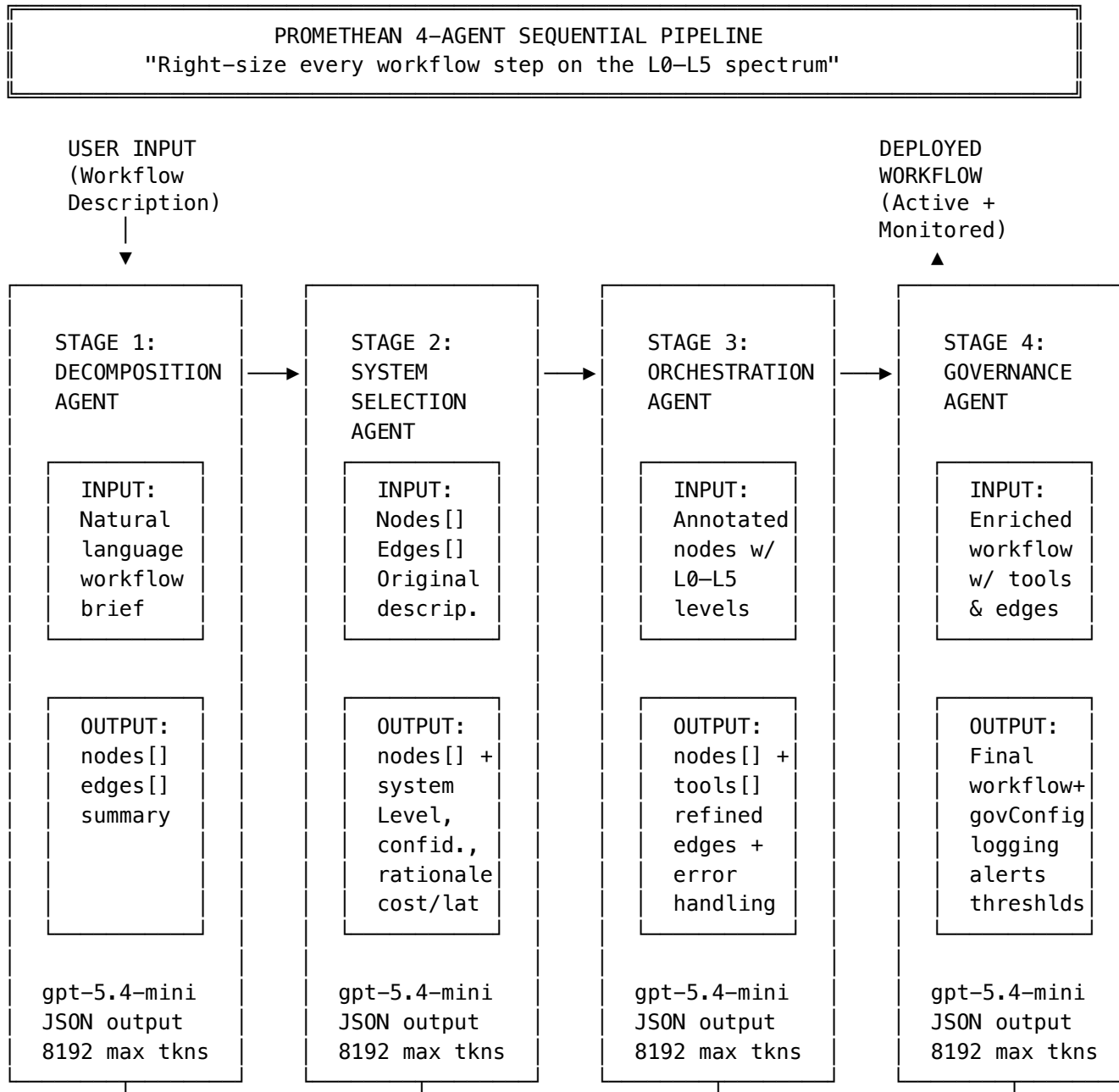
- workflowId + version
- definition (JSONB)
- changelog

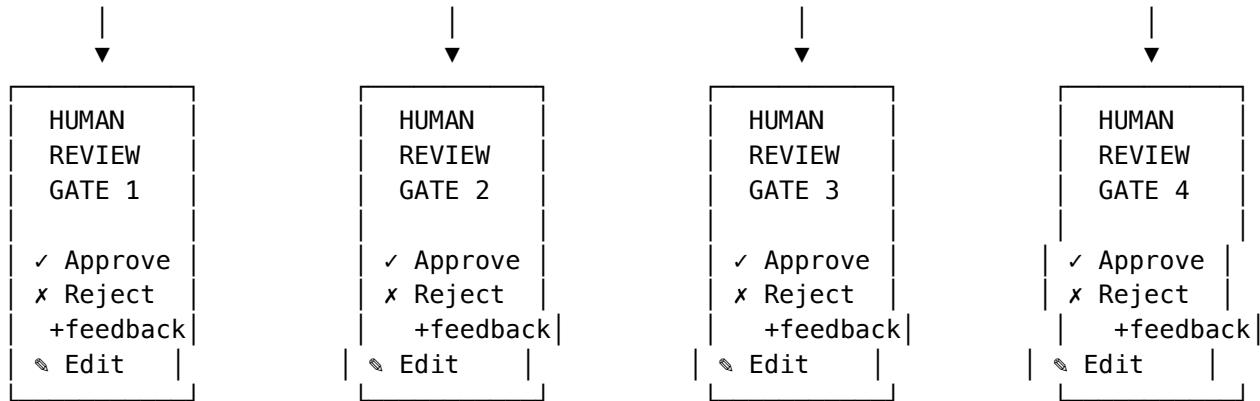
execution_steps

- executionId + stepId
- systemLevel
- input/output (JSONB)
- llmCalls, toolCalls
- cost, latency

2. Four-Agent Pipeline — Detailed View

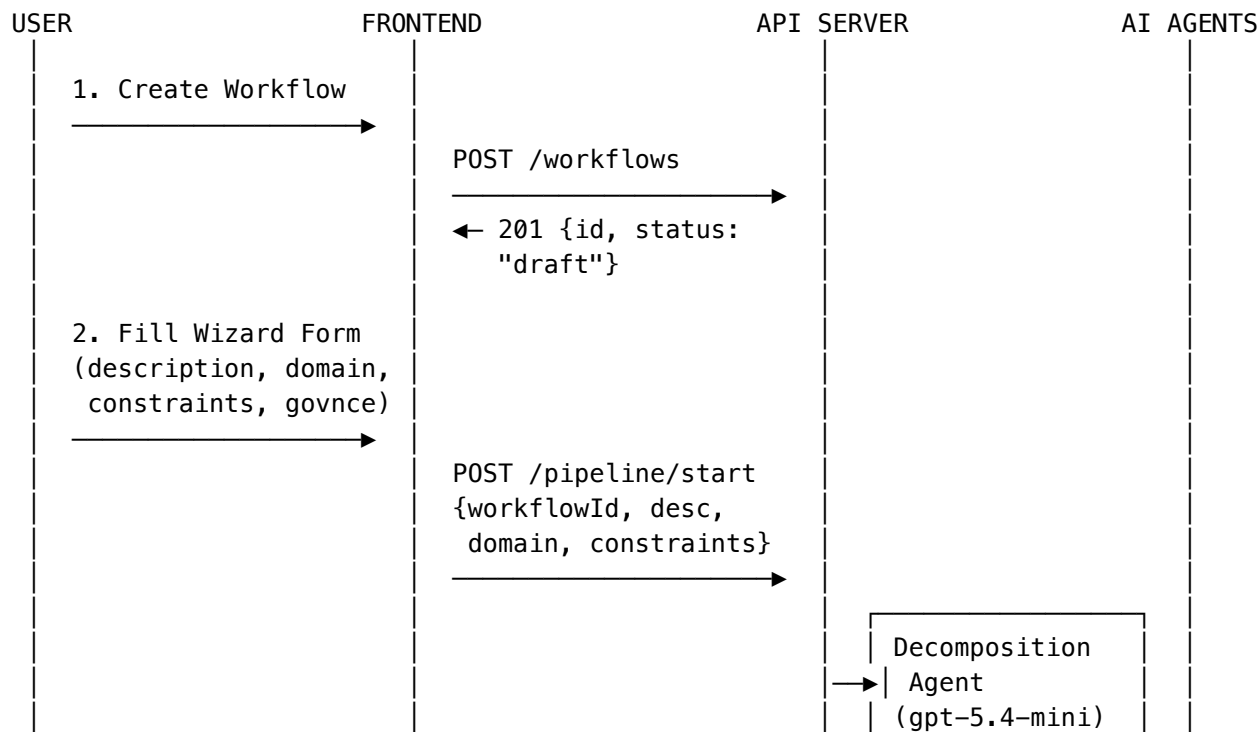
Model reference: gpt-5.4-mini in the ASCII diagrams below refers to gpt-5.4-mini-2026-03-17, the exact OpenAI model identifier used by all four agents in production code.

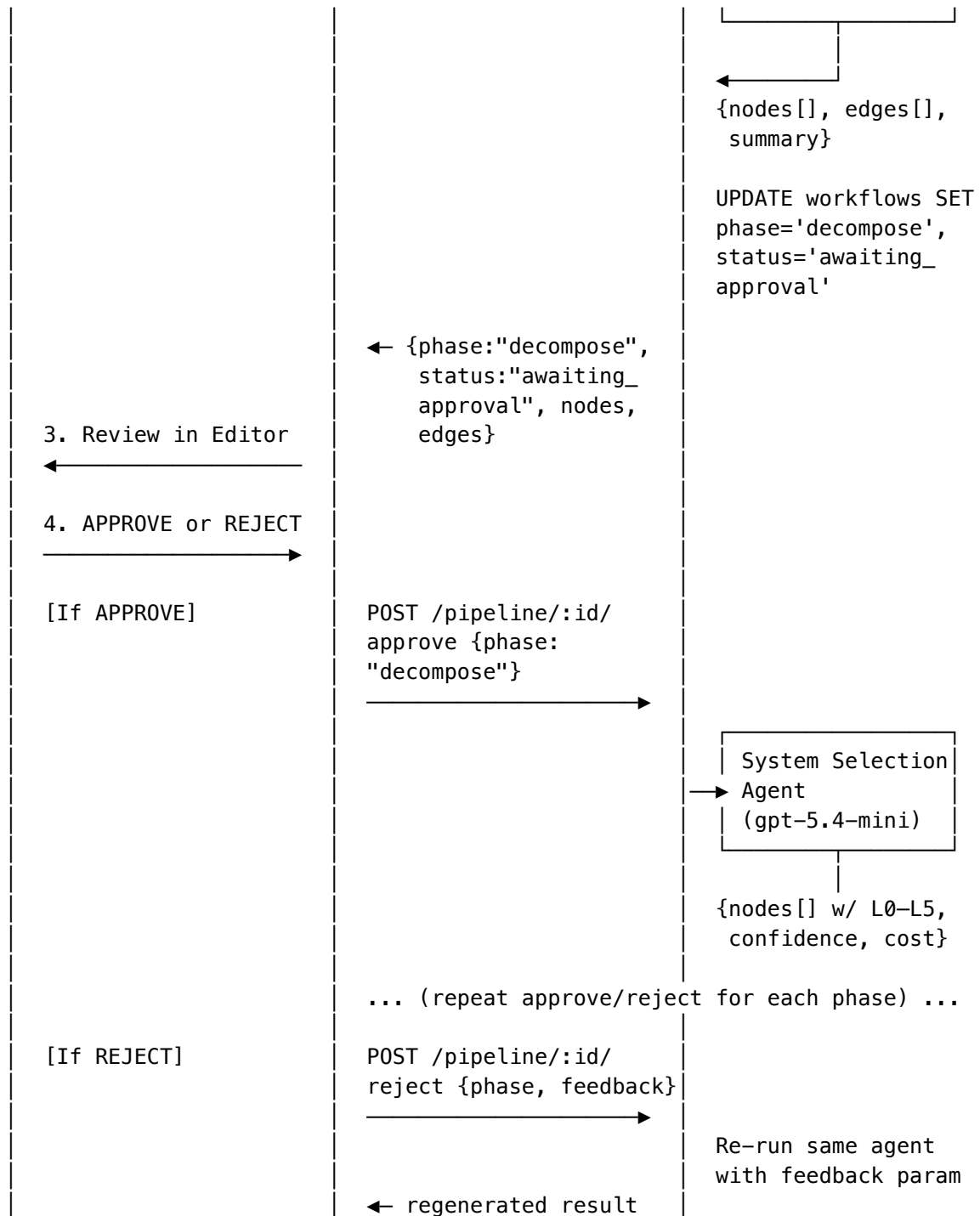


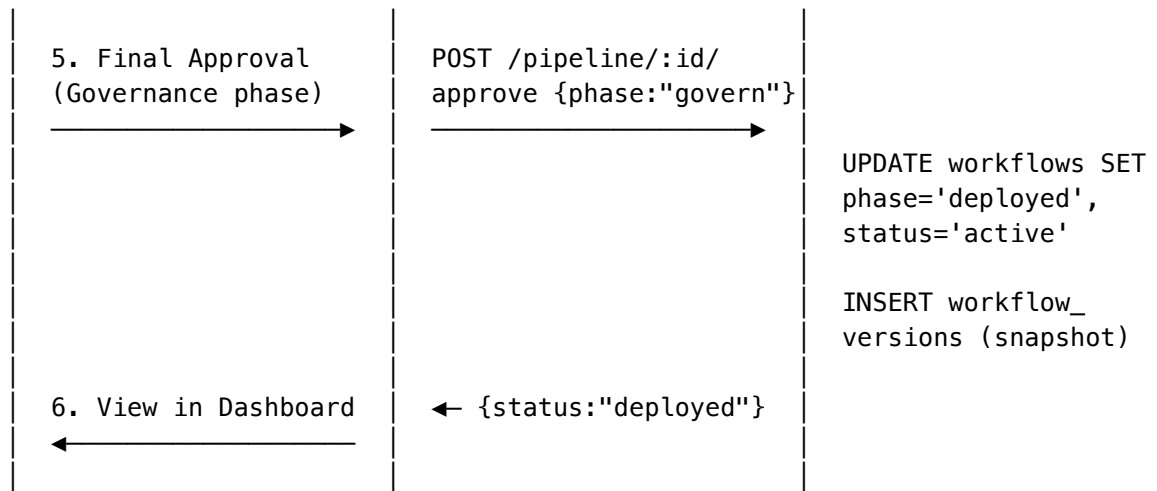


3. Request Flow — Complete Pipeline Walkthrough

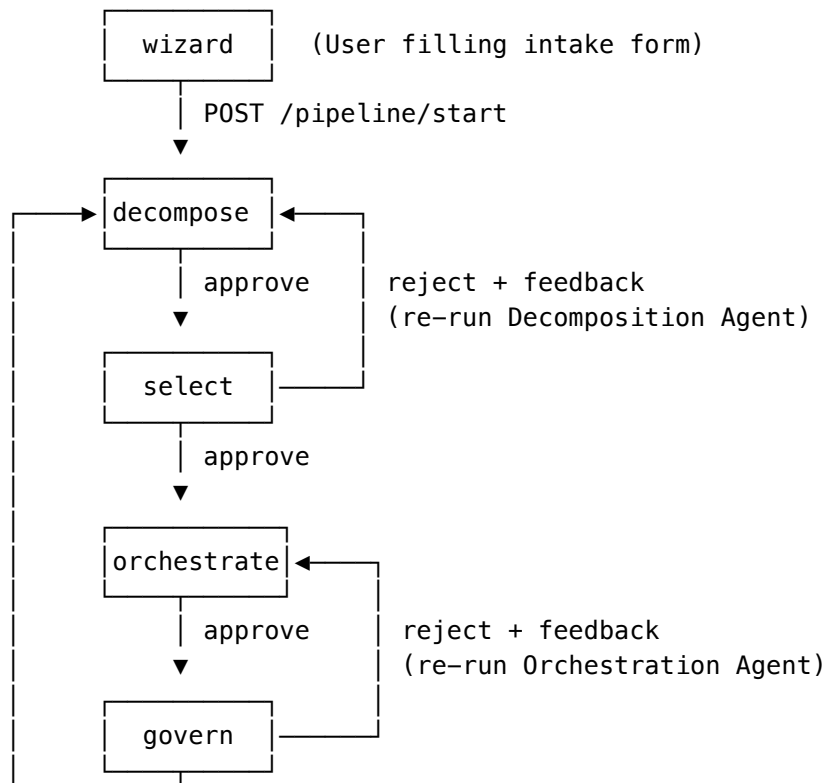
REQUEST FLOW: Workflow Through the Full Pipeline

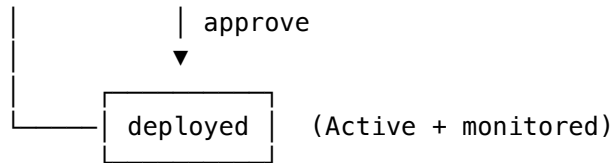






4. Pipeline Phase State Machine





5. Technology Stack Summary

TECHNOLOGY STACK
FRONTEND — React 19 + TypeScript — Vite (build tool) — Tailwind CSS + ARWES-inspired sci-fi theme — React Flow (visual workflow editor) — Recharts (data visualization) — Zustand (client state management) — React Query / TanStack Query (data fetching) — Orbitron + JetBrains Mono (typography)
BACKEND — Express.js + TypeScript — Pino (structured logging) — Zod (request/response validation) — CORS middleware
AI / AGENTS — OpenAI GPT-5.4-mini (all 4 agents) — JSON response format (response_format: json_object) — 8192 max completion tokens per call
DATA — PostgreSQL (primary data store) — Drizzle ORM (type-safe schema + queries) — JSONB columns (nodes, edges, configs)
INFRASTRUCTURE — Replit (hosting + deployment)


```
├─ pnpm monorepo workspace
├─ Shared packages (@workspace/db, @workspace/api-zod)
```

6. Monorepo Package Structure

```
promethean-studio/
├─ artifacts/
│   └─ promethean/          # React PWA frontend
│       └─ src/
│           ├── components/  # UI components (Dashboard, Wizard, Editor, etc.)
│           ├── hooks/      # React Query hooks
│           ├── stores/     # Zustand state stores
│           └─ pages/       # Route pages
│       └─ package.json
│   └─ api-server/         # Express API backend
│       └─ src/
│           ├── lib/
│           │   └─ agents/  # 4 AI agent implementations
│           │       ├── decomposition.ts
│           │       ├── systemSelection.ts
│           │       ├── orchestration.ts
│           │       ├── governance.ts
│           │       └─ types.ts
│           └─ routes/
│               ├── pipeline/ # Pipeline orchestration routes
│               ├── workflows/
│               ├── templates/
│               └─ alerts/
│       └─ package.json
├─ lib/
│   └─ db/                  # Shared database package
│       └─ src/
│           └─ schema/      # Drizzle ORM schemas
│               ├── workflows.ts
│               ├── executions.ts
│               └─ alerts.ts
```

```
├── templates.ts
├── packages/
│   └── api-zod/           # Shared Zod validation schemas
└── pnpm-workspace.yaml    # Monorepo configuration
```